

you'll have a working data pipeline and — for the first time — a weekly report showing you which of your databases are quietly wasting money and which are quietly heading towards an incident. A minimum viable deployment requires just three components:

Metrics pipeline: Prometheus with `postgres_exporter` and `node_exporter` pointed at target instances, and VictoriaMetrics configured for five-week retention.

Weekly scoring job: A Python script that queries VictoriaMetrics and outputs a classification — initially using hardcoded heuristics, replaced with the trained ML model once sufficient labelled history is available.

Human review step: Recommendations delivered by email each Monday morning. No automated execution.

This phase validates the data pipeline and begins building the labelled training dataset that makes the model smarter over time. Automation is introduced incrementally — never all at once. Scale Down actions carry the lowest operational risk and are the appropriate first target: incorrect recommendations are reversible.

Scale Up automation follows with a lightweight approval gate. Full autonomous execution is appropriate once model accuracy demonstrates consistent reliability. The path from

human-reviewed reports to autonomous governance is measured in weeks, not months.

The data required to make intelligent infrastructure decisions already exists in every PostgreSQL deployment. The gap is the absence of a learning system that converts observability into governance action. The framework presented here closes that gap using open source tooling, a four-week observability window, and a classification model that improves continuously.

Infrastructure managed by this framework reduces waste, prevents incidents, and compounds in accuracy. It is a generalised blueprint for autonomous infrastructure governance — applicable to any workload that generates behavioural telemetry. The organisations that deploy this capability build infrastructure that earns the right to manage itself. **END** 🐧

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